VINAY KUMAR

Senior Software Engineer & System Architect · Full Stack (Node.js · NestJS · Angular)

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SUMMARY

Senior software engineer and system architect with 5+ years across high-scale multi-tenant travel SaaS and complex distributed systems / ERPs, covering system design, microservices and end-to-end delivery on Node.js, NestJS, Angular and TypeScript with MongoDB, PostgreSQL, Redis and AWS. At EasyRes (since 2022) I am the top contributor to a multi-tenant B2B/B2C travel marketplace — about 3,000 commits across 24 repositories and 480+ merged pull requests — where I work directly with client stakeholders from requirements to delivery, and have built the B2B agent portal, created the travel-insurance and Via.com ferry microservices, and integrated flight, hotel, ferry, transfer and payment suppliers. I also created OpsX (opsx.in), an industrial ERP and workforce OS, which I architected and run in production.

WORK EXPERIENCE

OpsX (opsx.in) — Opsx Apex Private Limited, Gaya Jan 2026 – Present

Principal System Architect & Product Creator

- Created OpsX, a multi-tenant industrial ERP and workforce OS (attendance, payroll, field staff, factory kiosk, procurement, stock) — 760+ commits as sole engineer; Nx monorepo of 7 apps over 50+ backend domains.
- **Edge computing:** React Native (Expo) field and kiosk apps; offline-first SQLite queue with batch sync; on-device face verification on kiosk tablets — no image leaves the device.
- **Tenant isolation & RBAC:** a shared or dedicated MongoDB database per tenant (by plan), a tenant-scope layer that pins every write, hierarchical entitlements and maker-checker approvals.
- **Automated payroll:** PF, ESI, TDS, professional tax, arrears and full & final, posted to a double-entry ledger in PostgreSQL through an outbox.
- **Gateway-free UPI collection** (built, pre-launch): UPI QR per invoice; the payer's UTR is a claim, not proof: compare-and-set approval, a one-use UTR index and bank-statement auto-matching (HDFC, SBI, Kotak, Axis).
- **Security & scale:** RS256 JWT with MFA, per-route rate limiting, helmet and CSP; Redis caching with version-stamped invalidation on the check-in and RBAC hot paths.
- **CI/CD:** GitHub Actions runs lint, Jest unit tests (700+ test files) and build on every PR; zero-downtime rolling deploys on AWS EC2 (two NestJS instances, BullMQ workers, Socket.IO, nginx).

EasyRes Travel Technologies Pvt. Ltd., Delhi Jul 2024 – Present

Software Development Engineer II (Full Stack)

Multi-tenant B2B/B2C travel marketplace: 20+ Node.js microservices, Angular 14 and white-label portals, subscriber, agency and agent tiers. Integrations: **Travelport and Sabre GDS**, **Atlas LCC aggregator**; **RateHawk, GRN Connect, Dida, Bedsonline, Via Hotels**; **2GO, Barkota, Starlite, Via** ferries; **Jayride** transfers; **Xendit, PayMongo**; in-house packages, tours and visa.

- **Client-facing ownership:** gathered requirements with client stakeholders and designed solutions for ferries, hotels, visa, packages, airport shuttles and payments; owned delivery end to end.
- **Round trips on one-way supplier APIs:** Barkota and 2GO lack native round trips, so designed them on single-journey APIs: separate searches priced together, one payment, two reservations tracked per leg. As 2GO cannot cancel or release seats, both legs are validated before any hold.
- **Ferries:** built the **Via.com ferry service** from scratch (seat block, payment, restart-safe status polling); 2GO's B2C API with restricted fares and per-agency permissions; Barkota's B2B channel; Starlite refunds, voids and ticketing.
- **Money-safe bookings:** atomic claims before wallet or supplier side effects, automatic wallet reversal, manual review for unknown outcomes; proven by a **138-check offline test harness** and a ferry regression suite.
- **Hotels:** main contributor to **RateHawk**, built from client requirements (booking, status, cancellation penalties, hourly S3 log archive); Via Hotels per-tenant credentials and pricing fix.
- **In-house products (no third-party supplier):** visa processing modules plus the original visa service setup; holiday and tour packages with tenant-managed suppliers, prices and currencies (140+ backend commits).
- **Shuttles & insurance: Jayride** booking, wallet payment and redirects to client use cases; **created the travel-insurance microservice** (quotes, policies, gateway and wallet booking).
- **Payments & pricing: Xendit** end to end (sessions, webhooks, per-tenant config), PayMongo checkout, email-OTP login; grade codes, agency markups, multi-currency precision; Angular flight booking over Travelport, Sabre and Atlas.
- **Engineering practice:** Agile delivery tracked in Jira; merged 480+ pull requests incl. release and hot-fix branches; mentor to junior developers; Claude Code and GitHub Copilot, with every change reviewed and tested.

EasyRes Travel Technologies Pvt. Ltd., Delhi Jun 2022 – Jul 2024

Frontend Lead / Full Stack Engineer

- **Top contributor** to the Angular consumer portal and the subscriber-admin portal (about 280 admin components across flights, hotels, tours, visa, ferry, shuttle and insurance).
- **Built the B2B agent portal from scratch** (Nov 2023): flights, hotels, account management, agent home page.
- Built the B2C hotel booking flow (search, filters, paging, cancellation policy, promo codes), later Hotel v2 for B2B/B2C.
- Built airport-transfer and visa modules (search, application, status tracking); set up the visa microservice in Node.js.
- Built multi-method checkout with per-method absolute or percentage surcharges (card, bank transfer, UPI, e-wallets), wallet top-up and credit requests, agency earnings, grade plans and agency/airline markup screens.
- Created a per-product SCSS theming system used by several white-label tenant portals; booking voucher, itinerary and e-ticket generation.

VKREATE Software Technologies Pvt. Ltd., Jaipur Jun 2021 – Jun 2022

Junior Front-End Developer

- Built and maintained production CRM dashboards in Angular; integrated REST APIs for real-time data visualization, reporting.
- Built responsive web pages and UI components with React.js, JavaScript, HTML, CSS, Bootstrap and SASS; improved UI performance and usability.

VKREATE Software Technologies Pvt. Ltd., Jaipur · *Web Development Intern (JavaScript)* Apr 2020 – Dec 2020

- Completed a nine-month internship in web development with JavaScript, working as part of the development team.

OYS InfoTel Pvt. Ltd., Jaipur · *Web Development Intern (PHP/MySQL)* Jun 2019 – Jul 2019

- Worked on Educationwork.com, doing web development with PHP and MySQL, trained under a senior web developer.

SELECTED PROJECTS

Clinic Queue — real-time token & queue SaaS

Express 5, TypeScript, PostgreSQL, Socket.IO, React 19, Capacitor

- Postgres transactional outbox (LISTEN/NOTIFY + SKIP LOCKED workers) driving WhatsApp/SMS alerts with expiry, state re-check and SMS fallback; live queue; Android app; deploy kit with rollback.

SafeStep — personal safety app

NestJS, MongoDB, BullMQ, Redis, Socket.IO, Expo

- Server-side escalation on a missed timed check-in (BullMQ delayed jobs); live location to trusted contacts; 27 integration tests.

Content Automation Platform — video, podcast & social publishing (personal)

Python, FFmpeg, Node.js, YouTube Data API, Meta Graph, LinkedIn & X APIs, Cloudflare R2

- Resumable multi-stage pipeline (script, voice, images, render, scheduled upload) with per-stage checkpoints and a cross-process ledger lock; idempotent fan-out to YouTube, Instagram, Facebook, LinkedIn and X via a dedup ledger with atomic writes; self-hosted podcast RSS on R2; 65 tests.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3 / SCSS, SQL

Backend & Architecture: Node.js, NestJS, Express, TypeScript, REST & SOAP APIs, microservices, multi-tenant SaaS, RBAC, event-driven, message queues (BullMQ, Postgres outbox), JWT auth & MFA, rate limiting, idempotent payment flows

Frontend & Mobile: Angular, RxJS, Angular Material, React, Next.js, React Native (Expo), Capacitor, SCSS theming, Tailwind

Data: MongoDB / Mongoose, PostgreSQL, Redis (caching), BullMQ, SQLite (offline sync)

Cloud & DevOps: AWS (EC2, S3, SES), Docker, nginx, Linux, CI/CD (GitHub Actions), Nx monorepo, zero-downtime rolling deploys, cron jobs, Git

Testing & AI tooling: Jest unit testing, offline test harnesses (in-memory DB, mocked suppliers), integration and regression suites, Claude Code, GitHub Copilot

Domain: travel booking (flights via GDS and LCC aggregators, hotels, ferries, airport transfers, packages, insurance, visa), supplier API integration, payment gateways (Xendit, PayMongo) & wallets, multi-currency pricing, markups & grade codes, agency hierarchies, attendance & payroll (PF, ESI, TDS)

Leadership & delivery: client requirements gathering, stakeholder management, solution design, Agile delivery (Jira), end-to-end feature ownership, code review, mentoring

EDUCATION

Bachelor of Technology (B.Tech) — Computer Science Engineering, NIMS University, Jaipur 2021